

De-colonizing herring fishery models

June 4, 2015

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As an ecologist working for Central Coast Nations, I live at the interface of two worlds. On the one hand, I know that computer models can be important management tools. On the other hand, my job constantly reminds me that whether a model actually improves fishery management depends, fundamentally, on the worldview that shapes the model's objectives. To explore why, I will first explain some general concepts about what models can and cannot do. After that, I will summarize a recent model of herring populations and then pull it all together in a way that is relevant to First Nations.

Models do a great job of distilling the essence of how an ecosystem *might* respond to external forces—such as fisheries—but only under the specific conditions that the modeller assumes to be true in the “world” of the model. Sometimes these assumptions are well-grounded in reality. Sometimes they are blatant but necessary simplifications. Otherwise, it would be difficult to ask questions about how major forces for which we have no historical precedent—such as the combined effects of industrial fisheries, ocean acidification and climate change—might be altering the ocean. For instance, due to our greenhouse gas emissions, the ocean is warming and contains less dissolved oxygen. These stressful conditions hamper the capacity of fish to grow, and appear to be on their way to shrinking the body sizes of entire fish communities¹. If you want to even begin to comprehend what the ocean will look like in the long term due to these effects of climate change, it makes sense to assume, in the “world” of your model, that fishing does not exist, even though you know it does. Of course, you would then acknowledge that climate change probably exacerbates the effects of fisheries, which highlights that you still have to examine the combination of these effects. And that is exactly what an excellent team of modellers did¹.

So, as long as everybody is clear on the assumptions, models can provide useful guidance on how to manage fisheries. Of course, good fishery management would not rely solely on models but, as much as possible, would also use observations from the real world to verify and improve the models. It would be nice to rely on real world observations alone, but the combined forces of industrial fisheries, rapid climate change, and ocean acidification now occur at a speed and scale for which we have no precedent. In short, as imperfect as they are, we cannot afford to ditch models.

At the same time, we have to keep in mind that while the methods that scientists use to answer questions are objective, the individual worldviews that ask these questions are not. For instance, if a modeller works in isolation from indigenous people, chances are that the model he or she produces will not be designed to address questions relevant to indigenous people.

Let's switch modes and talk herring. DFO, like fishery management agencies elsewhere, has a long history of using models to guide herring management. I am not an expert on these models, but I am reasonably aware of the latest one that is about to be released publicly². After being exposed to that model during a recent two-day meeting, I

am impressed by it. As far as I can tell, it is technically sound and transparent with its assumptions. No arguments there. The catch is that the model assumes, implicitly, only one type of herring fleet: one that targets spawning adults for roe. Fair enough. The scientists who created the model were given a specific task by DFO fishery managers. The fundamental problem is that this task was constrained by an institutional worldview with a long history of ignoring First Nations.

Now, let's consider an alternative model of herring fisheries that emerged after a scientist opened his worldview to First Nations concerns. This is where Phil Levin comes into the picture. Phil is a lively character, a brilliant fishery ecologist who runs the Ecosystem Science section of the Northwest Fisheries Science Center (A US federal agency). The cool thing about Phil, who I have met personally several times, is that he will tell you right off the bat that "models are bogus". It is only after pausing to ensure that you heard what he said that he will back up a bit and elaborate on the sorts of things I have been talking about: assumptions, worldviews, visualizing potential drivers of change, and so forth. What Phil is trying to tell us, I think, is that that you have to open your mind to create useful models. Leading by example, Phil took leave from his job and went with his family to hang out in Haida Gwaii for a big chunk of 2010: seeing the territory first hand, getting sea sick while trying to catch salmon (I was not there, but have credible witnesses) and—critically—connecting with the Haida and learning about their historical and current fisheries.

Once back home, Phil gathered his team of modellers—Andrew Olaf Shelton, Jameal Samhour, and Adrian Stier—told them what he learned in Haida Gwaii, and set them loose to crunch numbers. The result is...well...another damn model of herring fisheries³! But hey, this one is different because it contrasts fisheries that harvest only eggs (the dominant First Nation fishery for commercial or FSC purposes) against fisheries for spawning adults (the main commercial fishery outside First Nations). The upshot of the model is that fishers can harvest a much higher proportion of eggs than of adults without having significant impacts on the herring population. Accordingly, harvest control rules should be much more conservative for the adult than for the egg fisheries.

This general result is consistent not only with what First Nations already know, but also with what ecologists already understand about animal populations. That is, very few eggs survive to become adults, so harvesting lots of eggs removes only a small proportion of individuals which are likely to reproduce in the future. In contrast, fishing for spawning adults directly removes the fish that are about to contribute new offspring to the next generation.

Let's look into some specific highlights of the model. In the simple case in which only one type of fishery occurs at the time, the model predicts that fishing 50% or more of spawning adults is likely to cause population declines that require a fishery closure. In contrast, the herring population changes little with up to 70% harvest of eggs. When both fisheries occur at the same time, any increase in the adult fishery severely reduces the number of eggs, but it takes much larger increases in egg harvest to reduce the number of adults.

Independently of fisheries, the abundance of herring is influenced by variation between years in recruitment, which is the number of juveniles that survive past their earliest, most vulnerable stages and become new members of the population. This variation might occur if juveniles encounter particularly unfavourable conditions for

survival—such as less food, more predators, or stressful temperatures—during some years but not others. According to the model created by Phil’s team³, when recruitment is very variable over time, there is a higher chance that a worrisome population decline will occur, even if there is no fishing of any kind. The implication is that, when recruitment varies a lot, restrictions on the adult fishery need to be ramped up sooner and more stringently than for the egg harvest. If herring biomass continues to decline, however, then egg harvests must close too.

Finally, model predictions go on to suggest that—when recruitment does not vary strongly between years—high rates of egg harvest not only are sustainable but also leave lots of extra herring for salmon, lingcod, whales and other predators to feed on. In contrast, high rates of fishing for adults are less likely to leave enough herring to support other species.

To be fair these are only model predictions and, as in all modelling studies, the authors make lots of simplifying assumptions³. Still, the assumptions are clearly laid out, and my version is that they are a more realistic representation of a world relevant to First Nations, other users and the ecosystem as a whole than I had seen in prior herring models.

As logical as the conclusions of the study may sound to First Nations and many ecologists, the simple idea of separating the impacts of adult fisheries and of egg harvest had not been applied before to a fishery model for Pacific herring. This goes to show how the worldview of the modeller plays a big role in determining the extent to which the model might address the concerns of First Nations.

The key issue is whether we can break past the institutional inertia that has constrained DFO herring fishery models inside a colonial worldview. I do not have a crystal ball, but my guess is that positive change might be possible. During the recent meeting with DFO about the latest herring model², Sean Cox, an SFU professor and fishery scientist who was contracted by DFO to lead the work, was receptive to the concerns and views expressed by the many First Nation representatives in the room. Perhaps the discussion that day was early first step towards de-colonizing herring fishery models.

References

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